

AIIMS CRE 4 (2025) Memory-Based Questions

Post Code 11: Anesthesia Technician / OT Technologist

Q1. Which drug controls bleeding in a pre-eclampsia patient undergoing LSCS?

Answer: Oxytocin

Explanation: Methergine is strictly contraindicated in pre-eclampsia because it causes severe vasoconstriction and hypertensive crises. Oxytocin is the preferred safe alternative for uterotonic effect.

Q2. Where is ETCO₂ measured in the breathing circuit?

Answer: Expiratory limb / Near mask

Explanation: ETCO₂ (End-Tidal Carbon Dioxide) is optimally measured near the patient's airway connection (at the Y-piece or expiratory limb) to capture exhaled alveolar gas accurately.

Q3. What is the Capnograph (ETCO₂) shape in COPD or Obstructive lung disease?

Answer: Prolonged expiratory phase (Shark-fin appearance)

Explanation: Due to bronchospasm and delayed alveolar emptying, the expiratory upstroke on the capnograph is slanted and prolonged.

Q4. What is the normal shape of a capnograph waveform?

Answer: Square form (Rectangle/Square wave)

Explanation: A normal capnogram has a sharp upstroke, a flat alveolar plateau, a sharp downstroke, and a baseline at zero, resembling a square wave.

Q5. What is the ideal method to detect an ET tube cuff leak?

Answer: Decrease in delivered tidal volume

Explanation: While airway pressure may drop, a discrepancy between the ventilator's set tidal volume and the exhaled tidal volume measured is the most accurate indicator of a cuff leak.

Q6. Which induction drug causes both profound hypotension and bradycardia?

Answer: Propofol

Explanation: Propofol causes significant vasodilation and myocardial depression. It also blunts the baroreflex response, which can lead to bradycardia alongside the expected hypotension.

Q7. Which device is used for High FiO₂ delivery?

Answer: HFNC (High Flow Nasal Cannula) / Non-Rebreather Mask

Explanation: High Flow Nasal Cannula or Non-rebreather masks are designed to deliver close to 100% oxygen (FiO₂ near 1.0) to spontaneously breathing patients.

Q8. Which IV fluid is contraindicated (not used) in head injury?

Answer: Dextrose 5% (D5W)

Explanation: Dextrose solutions become hypotonic once the glucose is metabolized. Free water then crosses the blood-brain barrier, exacerbating cerebral edema and increasing intracranial pressure.

Q9. Which fluid is NOT used in a sepsis patient?

Answer: Colloids (e.g., Hetastarch)

Explanation: Starches/colloids are associated with an increased risk of acute kidney injury and mortality in critically ill septic patients. Crystalloids (RL, Plasmalyte) are preferred.

Q10. Which complication increases the risk of awareness under anesthesia?

Answer: Hypotension

Explanation: If a patient develops severe hypotension, the anesthetist may intentionally reduce the depth of anesthesia to preserve blood pressure, inadvertently risking intraoperative awareness.

Q11. What is the color of a CO₂ cylinder?

Answer: Grey

Explanation: According to international safety standards, medical carbon dioxide cylinders are painted uniformly grey.

Q12. What is the color of a Nitrous Oxide (N₂O) cylinder?

Answer: Blue

Explanation: Nitrous oxide cylinders are standardized worldwide with a blue color.

Q13. What is the Pin Index Safety System (PISS) position for Oxygen?

Answer: 2, 5

Explanation: The PISS configuration for oxygen is pins 2 and 5. This prevents oxygen cylinders from being erroneously attached to other gas yokes.

Q14. Which device in the anesthesia machine regulates fresh gas flow?

Answer: Flow control valve

Explanation: Flow control valves on the rotameter assembly act to precisely control and regulate the release of fresh gases into the machine's manifold.

Q15. What is the function of the scavenging system?

Answer: Remove excess waste gases

Explanation: It safely captures and expels volatile anesthetic waste gases from the OR to prevent occupational exposure to healthcare workers.

Q16. What is the main function of Soda Lime?

Answer: CO2 Absorption

Explanation: In a closed or semi-closed circle circuit, soda lime chemically absorbs exhaled carbon dioxide, allowing the safe rebreathing of anesthetic gases and oxygen.

Q17. How is the size of a Guedel (Oropharyngeal) airway measured?

Answer: From the corner of the mouth to the angle of the mandible

Explanation: This ensures the airway sits over the tongue properly without pushing the epiglottis closed or falling short of clearing the base of the tongue.

Q18. Which ET tube is preferred for upper cleft lip/palate surgery?

Answer: RAE (Ring-Adair-Elwyn) tube

Explanation: The RAE tube has a preformed anatomical bend that directs the external portion of the tube downwards (oral RAE), keeping it out of the surgical field on the face.

Q19. Which device definitively prevents pulmonary aspiration?

Answer: Cuffed ET tube

Explanation: An inflated cuff on an endotracheal tube seals the trachea below the vocal cords, providing a definitive barrier against gastric contents entering the lungs.

Q20. Which device is used to examine the airway for intubation?

Answer: Laryngoscope

Explanation: A laryngoscope consists of a handle and blade used to retract the tongue and epiglottis, offering a direct line of sight to the vocal cords.

Q21. Which airway device can be introduced blindly WITHOUT a laryngoscope?

Answer: LMA (Laryngeal Mask Airway)

Explanation: Supraglottic airways like the LMA are designed for blind insertion into the hypopharynx, seating over the laryngeal inlet without requiring direct visualization.

Q22. What material is a standard laryngoscope typically made of?

Answer: Steel (Stainless Steel)

Explanation: Reusable laryngoscope blades are constructed from surgical-grade stainless steel to endure rigorous cleaning, autoclaving, and applied forces.

Q23. Which drug is NOT used in Deep Vein Thrombosis (DVT)?

Answer: Protamine Sulfate

Explanation: Heparin (LMWH or Unfractionated) is used to *treat* DVT. Protamine sulfate is the *antidote* for heparin, which would counteract the required anticoagulation.

Q24. Sugammadex is NOT used to reverse which muscle relaxant?

Answer: Atracurium

Explanation: Sugammadex is highly specific for steroidal muscle relaxants (Rocuronium, Vecuronium). It cannot bind to benzylisoquinolinium drugs like Atracurium.

Q25. What is the recommended oral suctioning pressure for adults?

Answer: 100 to 150 mmHg

Explanation: This pressure range is strong enough to effectively clear thick secretions while preventing mucosal damage, bleeding, and vagal stimulation.

Q26. Which of the following is NOT used in Rapid Sequence Intubation (RSI)?

Answer: Methylene Blue

Explanation: RSI uses fast-acting induction agents (Propofol) and fast-acting relaxants (Succinylcholine, Rocuronium). Methylene blue has no role in RSI; it's used for methemoglobinemia.

Q27. Which drug is an Induction agent?

Answer: Ketamine

Explanation: Ketamine, Propofol, Thiopentone, and Etomidate are standard intravenous induction agents. Ketamine provides profound analgesia and preserves airway reflexes.

Q28. Which Local Anesthetic contains an Ester linkage?

Answer: Procaine

Explanation: Local anesthetics are classified into Amides (Lidocaine, Bupivacaine - have two 'i's in the name) and Esters (Procaine, Tetracaine, Chloroprocaine - one 'i').

Q29. Which breathing circuit is used for a child weighing less than 10 kg?

Answer: Jackson-Rees (JR) circuit / Mapleson F

Explanation: The Mapleson F (Jackson-Rees) circuit provides very low resistance and minimal dead space, making it ideal for spontaneously breathing infants and small children.

Q30. Which semi-closed circuit is used for Adults?

Answer: Bain circuit (Mapleson D)

Explanation: The Bain coaxial circuit is the most efficient Mapleson circuit for controlled ventilation in adults, preventing rebreathing if adequate fresh gas flow is supplied.

Q31. What is the most common complication of a Subarachnoid (Spinal) block?

Answer: Hypotension

Explanation: Spinal anesthesia blocks preganglionic sympathetic nerves, causing vasodilation, venous pooling, and subsequent significant hypotension.

Q32. In a 3-lead or 5-lead ECG, where is the RED lead typically applied?

Answer: Right Arm (RA)

Explanation: According to standard AHA color coding, White is Right Arm, Black is Left Arm, Red is Left Leg. (However, in some international IEC coding, Red is Right Arm). The candidate noted right arm based on IEC standards.

Q33. Where is the most commonly used site for SpO2 probe placement?

Answer: Finger

Explanation: The fingers are highly vascularized, easily accessible, and provide reliable pulsatile waveforms for pulse oximetry calculation.

Q34. What is the most common artery used for Invasive Blood Pressure (IBP)?

Answer: Radial Artery

Explanation: It is superficial, easy to palpate/cannulate, and benefits from collateral blood flow to the hand from the ulnar artery.

Q35. Which ligament is crossed last when performing a subarachnoid block?

Answer: Ligamentum Flavum

Explanation: The needle passes through the skin, subcutaneous tissue, supraspinous ligament, interspinous ligament, and finally the thick ligamentum flavum to enter the epidural/subarachnoid space.

Q36. Hyperkalemia treatment includes all EXCEPT:

Answer: Central line placement

Explanation: Treatments for hyperkalemia include Dextrose + Insulin, Calcium Gluconate, and Salbutamol (which shift K⁺ into cells). Central line placement is not a direct treatment for hyperkalemia.

Q37. Which of the following is NOT part of preparation for an obese patient?

Answer: Long-acting drugs

Explanation: In obese patients, you prepare a difficult airway cart and check the OR table weight limit. Long-acting drugs are avoided because lipophilic drugs accumulate in fat, causing delayed recovery.

Q38. What is the most appropriate method to confirm proper ET tube placement?

Answer: Capnography (ETCO₂)

Explanation: While auscultation and chest rise are initial clinical signs, sustained ETCO₂ waveforms over several breaths are the gold standard definitive proof of tracheal intubation.

Q39. What is the CPR compression-to-ventilation ratio for adults?

Answer: 30:2

Explanation: AHA guidelines recommend 30 chest compressions followed by 2 rescue breaths for adult basic life support.

Q40. What action leads to CPR failure / poor outcomes?

Answer: Gap (interruption) of more than 30 seconds

Explanation: AHA stresses minimizing interruptions in chest compressions to less than 10 seconds. Interruptions of 30 seconds cause coronary perfusion pressure to drop to zero.

Q41. What is the first drug given during CPR (cardiac arrest)?

Answer: Adrenaline (Epinephrine)

Explanation: Adrenaline 1 mg every 3-5 minutes is the first-line vasopressor in all ACLS cardiac arrest algorithms to improve aortic diastolic pressure.

Q42. What is the specific reversal drug for Non-Depolarizing Muscle Relaxants?

Answer: Neostigmine

Explanation: Neostigmine is an acetylcholinesterase inhibitor that increases acetylcholine at the neuromuscular junction to compete with and displace the relaxant.

Q43. Which gas supports combustion in the operating room?

Answer: Oxygen (and Nitrous Oxide)

Explanation: Both Oxygen and Nitrous Oxide are oxidizers. While they are not flammable themselves, they fiercely support and accelerate the combustion of other materials (like drapes).

Q44. What is the minimum safe SpO2 range?

Answer: 95%

Explanation: A normal healthy adult should maintain an SpO2 of 95-100%. Below 90% is considered hypoxic and requires clinical intervention.

Q45. What is the MAC (Minimum Alveolar Concentration) value of Sevoflurane in adults?

Answer: 2.0%

Explanation: At 1 MAC (2.0% for Sevoflurane), 50% of patients will not move in response to a surgical incision.

Q46. Which is NOT a technique for a Brachial Plexus block?

Answer: Interscapular

Explanation: The approaches for a brachial plexus block are Interscalene, Supraclavicular, Infraclavicular, and Axillary. Interscapular is not a valid approach.

Q47. What is the primary function of an HME (Heat and Moisture Exchange) filter?

Answer: Maintain humidity

Explanation: HME filters conserve the patient's exhaled heat and moisture and return it during inspiration, preventing airway drying caused by cold, dry medical gases.

Q48. Which is NOT a part of the basic anesthesia machine?

Answer: Pulse Oximeter

Explanation: The basic Boyles machine consists of the yoke assembly, vaporizers, flowmeters, and CO2 absorbers. Pulse oximetry is a separate patient monitoring module.

Q49. The Aldrete Score is used for what purpose?

Answer: Discharge from PACU (Post Anesthesia Care Unit)

Explanation: It assesses activity, respiration, circulation, consciousness, and O2 saturation to determine if a patient is safe to leave recovery.

Q50. Which inhalational agent is an excellent Bronchodilator?

Answer: Sevoflurane (also Halothane)

Explanation: Sevoflurane is smooth, non-irritating to the airway, and causes significant bronchodilation, making it the agent of choice for asthmatic patients.

Q51. Which agent has the Minimum MAC value (highest potency)?

Answer: Halothane (0.75) or Isoflurane (1.15) depending on options.

Explanation: Potency is inversely proportional to MAC. The lower the MAC, the more potent the drug. Isoflurane (1.15) is highly potent compared to Sevo (2.0) and Desflurane (6.0).

Q52. What is the definitive means of airway management?

Answer: Endotracheal (ET) tube

Explanation: An ET tube passes through the vocal cords directly into the trachea and features a cuff, making it the only "definitive" airway.

Q53. What is the purpose of Pre-oxygenation?

Answer: To extend the safe apnea time

Explanation: Breathing 100% O2 washes out nitrogen from the functional residual capacity (FRC), creating a reserve that prevents desaturation during the apneic intubation attempt.

Q54. Boyle's law is related to which monitor graph?

Answer: Pressure-Volume loop graph

Explanation: Boyle's Law states that at a constant temperature, Pressure is inversely proportional to Volume ($P \propto 1/V$). This concept governs the mechanics of ventilation.

Q55. What happens when Soda lime is exhausted?

Answer: CO2 Accumulation (Hypercapnia)

Explanation: Once the chemical neutralizing capacity of soda lime is spent, exhaled CO2 is rebreathed by the patient, causing a rise in baseline ETCO2.

Q56. What is the capacity of an E-type Oxygen cylinder?

Answer: 660 Liters

Explanation: A standard E-size cylinder (often attached to the back of anesthesia machines) holds approximately 660 liters of oxygen at roughly 2000 psi.

Q57. What is the standard size of an ET tube connector?

Answer: 15 mm

Explanation: ET tube proximal connectors have a universal 15 mm outer diameter to attach seamlessly to standard breathing circuits and Ambu bags.

Q58. What is the standard flow rate of the Oxygen Flush valve?

Answer: 35 to 75 Liters/min

Explanation: The oxygen flush bypasses the vaporizers and flowmeters to deliver pure oxygen at high pressure (about 50 psi) at 35-75 L/min for emergencies.

Q59. In case of electricity failure, which part of the anesthesia machine still runs?

Answer: Mechanical Flowmeter

Explanation: Glass rotameter (mechanical flowmeter) tubes rely purely on the physical gas pressure from the cylinder/pipeline and a needle valve, needing no electricity.

Q60. Which of the following is a type of Colloid fluid?

Answer: Hetastarch

Explanation: Hetastarch, Albumin, and Dextran are large-molecule colloids that stay in the intravascular space. NS and RL are crystalloids.
